

Mobility, Stability, Antagonist Training

To maximize your potential and reduce injury risk, it's essential to develop stability before strength, and strength before power.

Here's where the business begins! This is the first of four exercise-packed chapters that together hold the potential to transform your physique, elevate your quality of movement, and markedly increase your strength, power, and endurance. A consistent effort to properly apply the one-hundred-plus exercises presented in these chapters will literally reshape your body and undoubtedly open up a whole new level of climbing performance.

The significance of the results you obtain, of course, will depend on a number of things, including your current level of conditioning and injury history (if any), your age and genetics, and the time you have to invest in training. Far more important, however, is your willingness to engage in a comprehensive training program and do the right things right. When I talk about the “right things,” it's not just about appropriate climbing-specific exercise selection (which is very important), but also about following through faithfully with the essential flexibility, mobility, stability, and antagonist exercises. For example, dedicated training on the wrist and scapular stabilizers—while not as sexy as, say, campus training—is absolutely essential as both a precursor and complement to the climbing-specific exercises. Break this guideline and, sooner or later, you will injure a finger, elbow, or shoulder—or perhaps all three!

The ubiquity of injured climbers (an alarming number needing surgery) is evidence of the prevalence of over-enthusiastic training, climbing too much (with too few rest days), and a failure to properly train arm and shoulder mobility and stability before engaging in high-end climbing and advanced climbing-specific training (such as fingerboard and campus board training). While certainly not a 100 percent preventative of injury, embracing the concept of “stability before strength, and strength before power” will vastly lower your injury risk and enable you to engage in a progressively more rigorous training program with each passing year.

Figure 6.1 presents mobility and stability as a necessary foundation to be built before progressing to strength training and, eventually, power training. Disciplined use of most of the exercises in this chapter will build a physiologically sound foundation from which you can safely execute the many climbing-specific exercises in chapter 8 and 9. Failing to regularly utilize the foundational exercises in this chapter, however, is tantamount to building your climbing physique on a foundation of sand—one that will likely fail you in a critical moment with a fateful pop, tear, or a more insidious gradual development of pain in your shoulder, elbow, or finger.

Okay, enough with my obligatory coach's warning—let's get to the details that you can put to work beginning today! First, I'll present the essential mobility and flexibility exercises that you should do each day at home, at the crag, or in the gym. The real meat of this chapter, however, are the sections

Figure 6.1 Pyramid of Training



A period of mobility and stability training is essential before embarking on serious strength training. Graduate to power training only after you've developed significant climbing-specific and stabilizer strength.

that detail the vital stability and antagonist exercises that every climber should do. Read on and you'll learn how to keep your arms and shoulders functionally sound and, hopefully, healthy throughout a long climbing career.

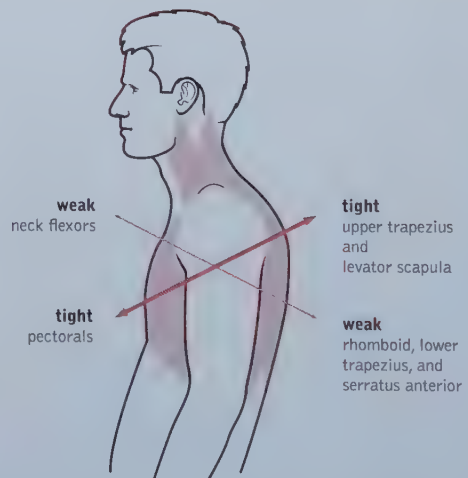
Flexibility and Mobility Training

First, let's distinguish between flexibility and mobility. Flexibility refers to the range of motion around a joint—for example, climbers need flexibility around the hips for high-stepping, stemming, and aggressive drop-knee positions. Flexibility is influenced by joint mechanics, tendon, fascia, and musculature, and in aggregate it's a system that will gradually respond to dedicated training. Mobility relates to functional movement patterns and how well a joint moves through the appropriate range of motion in specific physical activities. A joint's mobility is influenced by many factors including

motor control, chronic movement patterns, antagonist muscle strength, soft tissue adhesions and injury, and—yes—flexibility.

For climbers, the shoulder joint is a common site of poor mobility as tight, weak, sore, and/or injured muscles diminish mobility and compromise functional movement (see figure 6.2). The forearm muscles are another common site where weak antagonist muscles and soft tissue dysfunction can contribute to poor movement patterns, undue compensatory strain, and injury. Also common are mobility and flexibility issues surrounding the hips and lower back. Though often more a result of excessive time spent sitting at a desk or in a car, the long-term effects can impact your back, core, and hip function and therefore negatively impact your climbing. The bottom line: These common problem spots can be largely corrected given daily mobility and flexibility work.

Figure 6.2 Causes of “Climbers’ Posture”



Tight pectorals and weak lower trapezius muscles are two factors that often contribute to the rounded shoulders and rounding of the upper back—this increases the risk of shoulder impingement during forceful overhead reaching and pulling.

Upcoming are seven oh-so-good mobility exercises, nine upper-body warm-up and stretching exercises, and nine core and lower-body stretches. Invest fifteen to thirty minutes per day into doing a selection of these exercises and you'll be on your way to moving—and feeling—better, both on and off the rock! (Consult the muscular anatomy photos in appendix A if you are uncertain of the location of the muscles targeted in each of these exercises.)

Foam Rolling Exercises

There are many fitness fads that come and go each decade, and many are ineffective exercises or fitness equipment promoted heavily by manufacturers and marketers. The last few years have brought the rise of foam rollers and other tools for soft tissue management, and I can tell you with complete certainty that the benefits of these devices are entirely legitimate. I'm so enamored of my foam roller and Armaid device that I take them on all my climbing trips. These tools are real difference makers in terms of subjective feelings of muscle function as well as improved mobility.

Detailed below are several techniques to address tight muscles and, more importantly, hunt out and destroy painful trigger points. Each will provide self-myofascial release (SMR) that will help lengthen muscle fibers, release trigger points, increase circulation, and perhaps even encourage fascial remodeling. Make SMR a daily habit and you will improve mobility and lower the risk of certain overuse injuries such as medial and lateral epicondylitis.



UPPER BACK AND SHOULDER ROLL

Begin with the foam roller under your upper back and your arms across your chest. With your knees bent and feet flat on the floor, slowly roll up and down along the thoracic spine. Pause for fifteen to thirty seconds on any painful spots (trigger points) you locate in your trapezius and rhomboids. Next,



Foam rolling the upper back and shoulders with a Rumble Roller.

1. Targeting the trapezius, rhomboids, teres major/minor, infraspinatus, and posterior deltoids. 2. Rolling the latissimus dorsi.

tilt about 30 degrees to each side and roll along the latissimus dorsi and up across the posterior deltoid and infraspinatus. Again, dwell for a half minute or so on any sore or tight spots. Finally, return to the neutral position and extend your hands overhead. Keeping your elbows locked and maintaining butt contact with the floor, slowly roll up and down along the thoracic spine. This is excellent for improving thoracic extension and addressing the posture issues and shoulder strains common among climbers (as shown in figure 6.2).



LOWER BACK AND GLUTES ROLL

The muscles of the lower back and buttocks are common hiding spots of tightness among climbers. Anyone who spends large amounts of time sitting (commonly with a flexed lumbar spine) will find lower back rolling to be surprisingly therapeutic; it might even elicit a glorious “pop” (without dropping coin on a chiro!). Rolling lower onto the gluteal muscles of the buttock may be similarly

pleasurable. Many climbers discover extremely tight gluteus medius muscles (located on the sides of buttocks, just below the hip)—this isn't surprising, since it's the gluteus medius that produces external thigh rotation when the knee is flexed (as it is in just about every climbing move).

To address the lower back, position the foam roller perpendicular to the lumbar spine and roll slowly up and down along the often tight erector spinae muscles. Add a slight roll to each side and you'll likely find soreness in the quadratus lumborum (QL). The QL connects the pelvis to the spine, and it's a common source of lower back pain.

Rolling the gluteus maximus is best done from a sitting position with straight legs and arms extended behind you. To release the often tight gluteus medius, however, you'll want to bend one leg and then lean to the opposite side—roll just a few centimeters up and down and side to side and you'll quickly locate the trigger point. Dwell there for a minute and feel the tightness release.



1. Rolling the spinal erectors, quadratus lumborum, and gluteus medius.
2. Targeting the gluteus maximus.



HIP FLEXORS AND IT BAND

The hip flexors and iliotibial (IT) band are two more common tight spots to address with a foam roller. If you spend much time in the sitting position, then you'll surely discover that you have tight hip flexors that will hurt (in a good way) when addressed with a foam roller. Assume a prone position with the foam roller under your hips and only your toes and forearms contacting the floor. Now roll just a few centimeters up and down along the hip flexors (front of your upper thighs and pelvis). Tilt slightly to the side to attack the hip flexors only on that side—this will double the pressure (more painful), but yield even more of a release. Dwell for up to a minute on each group (side) of hip flexors.

Next, proceed to rolling the IT band along the outside of each leg. Lie on your side with the foam roller under the upper outside of one leg. You'll need to support much of your weight on the floor-side forearm and position the other leg for balance. Now slowly roll along the outside of the leg, tracking the roller from just below the hip to just above the knee.



1. Rolling the hip flexors and upper quadriceps.
2. Targeting the IT band.



PRONATOR “PIN AND STRETCH”

The pronator teres is but a small muscle in the upper forearm, but for a climber it's one of the hardest-working muscles. You can easily locate this muscle by extending an arm straight to the front and then making a tight fist while you pronate the hand completely (as if pouring water from a jug). The pronator teres will bulge on the inside of the elbow just above the bony medial epicondyle. Now relax your fist and, using the thumb of the opposite hand, press deeply into the pronator muscle (the “pin”) and hold for thirty seconds. Chances are you'll find this to be quite painful—a sign that you should also do the pronator strengthening exercise on page 136. Repeat the “pin” again with your thumb, but this time slowly pronate and supinate your hand to achieve a deeper massage and stretch of the pronator. Alternatively, use an Armaid on your pronator, as it provides even better leverage than using your thumb.



Pronate your hand to expose the pronator teres muscle (just above the medial epicondyle), then use your thumb to press deeply into this often tight muscle.



WRIST EXTENSORS “PIN AND STRETCH”

The finger and wrist extensor muscles of the lateral forearm are another common problem spot, especially among climbers who are chronic crimpers. Persistently tight extensor muscles often develop painful knots and trigger points at one or more locations from the middle lateral forearm up to within about one inch of the bony lateral epicondyle. While daily forearm stretching of the extensor muscles is essential (page 122), addressing these tight muscles with an Armaid will lessen tension, release trigger points, and ultimately reduce your chance of someday developing lateral tendinosis.

Here's how to do it: Using an Armaid (or your thumb), probe the soft tissue of the lateral forearm to locate any sore spots—when you find one, apply steady pressure and hold for thirty seconds. Next, “pin and stretch” this trigger point by making a fist and flexing your wrist as you apply increasing pressure on the sore spot. Climbers with numerous trigger points on the lateral forearm would be wise to invest at least five to ten minutes per day addressing these developing problem spots.



The Armaid is an excellent tool for addressing sore, tight trigger points in the finger and wrist extensor muscles.

As Essential as Your Climbing Shoes—Armaid!

We all know the feeling of sore, tight forearm muscles after a day of hard climbing or an intensive finger workout. If you climb and train frequently without taking proactive measures to address these tight muscles and trigger points, resultant muscle shortening will become chronic. Incessant tension of forearm flexor and extensor muscles produces an unwanted “pulling”—even during rest—where the muscles attach via tendon at the medial and lateral epicondyle of the elbow. Long term, this chronic irritation can eventually lead to either tendinitis (inflammation) or tendonosis (degradation of tendon collagen).

If you ever feel tendon pain near one of these epicondyles or in the attaching muscle an inch or two distal to the epicondyle, you now know what’s likely part (or all) of your problem. Even if you are pain free (congrats, but let’s keep it that way!), I’ll bet that you can elicit some painful twinges by firmly palpating deep into the muscles of the medial and lateral forearm. By proactively addressing these trigger points with sports massage and trigger point release, you’ll regain your natural range of motion and unlock a higher level of muscle function.

Here’s how to do it: With a braced finger or, better yet, an Armaid or Rubbit, hold firm pressure on a sore spot (trigger point) and then stretch the muscle. Feel the tension or trigger point release over the course of a minute or two of therapy. Locate other sore spots and similarly “pin and stretch” them until they submit. This can be a painful process, but you’ll quickly come to recognize and perhaps enjoy this kind of “good pain.” Of course, you mustn’t overdo it, or ever apply this technique directly to a tendon or epicondyle.

The bottom line: If you engage in stressful climbing or rigorous forearm training, then daily self-care of

the forearm muscles is essential for reducing tension and injury risk. Furthermore, a loose, relaxed muscle is a happy, fully functional muscle possessing optimal blood flow and highest recovery ability. Armaid and Rubbit are the two best massage tools I’ve used, and I consider them so critical for maintaining my forearm health that I take one on all my climbing trips. You should too!



The Rubbit excels equally as a tool for massage and trigger point therapy.



ELBOW VODOO FLOSS

Voodoo flossing is a novel technique for helping to mobilize tissues and increase circulation to cold, stiff muscles and joints, and it's been effective for improving tissue quality following a connective tissue injury. Voodoo flossing involves wrapping a tight rubbery band (available from RogueFitness.com) around a sore, stiff joint—in this case the elbow. Begin wrapping the voodoo floss about 4 inches below the elbow and tightly wrap, with 50 percent overlaps, all the way to about 4 inches above the elbow. Tuck the end of the band under the previous wraps. The goal is to wrap tight enough to occlude blood flow—your hand will turn purple or white—so shoot for a tightness of 8 on a scale of 1 to 10. Now flex and extend your elbow for about twenty slow repetitions, removing the voodoo floss after about one minute.



Improve circulation and tissue quality in aching elbows with regular voodoo flossing.

The effects of voodoo flossing come from the powerful reperfusion (return of blood flow) following removal of the band as well as the sliding and shearing of stiff tissues that results from the powerful compression the band provides. Long-term voodoo flossing on a chronically sore or previously injured elbow will help improve tissue mobility, increase range of motion, and enhance beneficial circulation. Also noteworthy, voodoo flossing is an excellent method of warming up the elbows before training or climbing.



PECTORAL RELEASE WITH LACROSSE BALL

The elbows and shoulders have been the focus of this section, and for good reason—next to the fingers these are the most common sites of injury among rock climbers. The last mobility exercise addresses what is, perhaps, the most overlooked muscle during climbers' warm-up and stretching



Many climbers have chronically tight pectoral muscles that can be readily addressed with a tennis or lacrosse ball. Here the author enjoys the good pain of some self-myofascial release.

routines. As shown in figure 6.2, tight pectorals contribute to the poor posture and shoulder dysfunction that many climbers exhibit. Releasing tight pectorals (and the upper trapezius) and strengthening the lower trapezius and rotator cuff muscles will go a long way to improving functional movement and reducing your injury risk.

While you can use a foam roller on the pectorals, a lacrosse ball or tennis ball (more forgiving) is a better match for this smaller muscle. You can best address the pecs by standing upright and placing the ball between your pectoral muscle and a wall or vertical column. You can easily move your torso position to direct the ball onto any tight spots. As with the previous exercises, pause with the ball on the sore spots and wait for the tissues to submit.

Upper-Body Stretches

While there is rarely a need for extraordinary flexibility in climbing, regular flexibility training will have a positive influence on your training and performance. Considering that movement is the very essence of the vertical dance we call climbing, anything you can do to help facilitate smooth, efficient movements will enhance your performance. Flexible agonist muscles and tendons will function better—and are more resilient—when exposed to the high dynamic force loads common to climbing. What's more, flexible antagonist muscles will levy a lower inherent resistance to the opposing agonist muscles, enabling smoother, more economical movement.

Detailed below are nine stretches that are ideal for fulfilling the above-stated goals.

Not surprisingly, stretching exercises have long been used by climbers as part of a warm-up routine before training, climbing, and competition. It's important to understand, however, that such pre-workout stretching must always be preceded by a warm-up activity that will increase muscle temperatures and spread synovial fluid through the joints and tendon sheaths. Static stretching alone can injure a cold muscle (Shrier 1999, 2000), so

it's essential to engage in five to fifteen minutes of low-intensity exercise such as jogging, easy climbing (vertical walls with good holds), or foam rolling before doing this flexibility work. When climbing outdoors, the hike to the cliff base often provides the perfect lead-in to your preclimb stretching as long as it's lengthy enough to increase your breathing rate and cause a light sweat.

At the very least, do twenty-five jumping jacks followed by twenty arm circles and twenty finger flexors (as described below). Complete the pre-stretching warm-up with a few minutes of self-massage to the finger tendons, the palm of your hand, and the forearm muscles and biceps.



ARM CIRCLES

Arm circles provide a nice dynamic stretch that warms up your shoulder joints and increases circulation to your shoulders, arms, and fingers. Do these before you engage in any of the upper-body static stretches that follow. Stand with your arms out to the side and parallel to the floor, and begin moving your arms in small circles. Gradually increase the size of the circle until you feel slight tension in the shoulders—go no larger with the circles beyond this point, and do not wildly whip your arms or perform a rapid windmill-like motion! Complete twenty smooth, moderately paced arm circles, and then proceed to the Finger Curls.



FINGER CURLS AND MASSAGE

This is a must-do warm-up movement for climbers because it increases circulation to the forearm muscles and spreads lubricating synovial fluid in the joints of the fingers. Stand with your arms relaxed by your sides. Close your hands to make relaxed fists, and then quickly open your hands and fan out the fingers as if you're flicking water off your fingertips. Continue for thirty to forty repetitions. Use a pace that allows about two repetitions per second. Now

perform a minute or two of massage on both sides of your fingers and hands—this will encourage blood flow and help warm the tendons and tendon pulleys.



FINGER ISOLATION STRETCH

This isolation stretch, along with massage of the fingers and hands, is very effective for warming up your precious digits. Work through this sequence one finger at a time. Either sitting or standing, position one hand palm-up at chest level and curl the fingers into a loose fist. Place the index finger of the other hand across the tip segment of the finger being stretched, and position the thumb under the finger near the base (hand) knuckle. Bracing with the thumb below, gradually apply pressure with the index finger to further close the bent finger and push it away from the palm. Stop when you feel mild tension in the joints. Hold the stretch for ten seconds. Release the finger for a few seconds, then repeat the stretch for another ten seconds. Repeat the stretching sequence with all eight fingers. Stop immediately if you experience any pain.



Finger isolation stretch



FINGER FLEXOR STRETCH

This rudimentary stretch targets the forearm muscles that flex your fingers and secure your grip to the rock. While in a standing position, bring your hands

together in front of your waist. Straighten the arm to be stretched and lay the fingertips into the palm of the other hand. Position the hand of the stretching arm so the palm is facing down with the thumb pointing inward. Pull back on the fingers of the straight arm until a mild stretch begins in the forearm muscles. Hold the stretch for ten seconds. Release the stretch and turn the hand 180 degrees so that the stretching arm is now positioned with the palm facing upward and the thumb pointing out to the side. Again, using your other hand, pull your fingers back until a stretch begins in the forearm muscles. Hold for ten seconds. Repeat the stretch, in both positions, with the other arm.



1. *Finger flexor stretch with hand pronated.*
2. *Repeat stretch with hand supinated.*



FINGER EXTENSORS STRETCH

This important, yet often overlooked, stretch targets the numerous extensor muscles of the lateral forearm as well as the commonly sore and tight brachioradialis muscle. These muscles are especially strained when crimping with a chicken-winged



Maintain a straight elbow while flexing the wrist and fully pronating the hand.

arm position, and so daily stretching (and Armaid use) is essential for lengthening the tissues and releasing tension that can eventually contribute to lateral epicondylitis.

With nearly straight arms, cross your hands in front of your body and interlace your fingers, palms together. While maintaining mild tension throughout the length of both arms, pull with one hand to flex the wrist of the other hand until you feel the stretch develop in the finger/wrist extensors along the outside of the forearm. Hold the stretch for about twenty seconds and then pull with the other hand to create a stretch along the other arm. Perform the stretch twice on each arm.



POSTERIOR SHOULDER STRETCH

This addresses the shoulder capsule and in particular stretches the infraspinatus and trapezius. While in a standing position, bring one arm across your chest until the hand rests on the opposite shoulder. Importantly, you must have your elbow bent (around 90 degrees) and maintain a shoulder back position—do not roll your shoulder forward. With the other hand, grasp behind the bent elbow from below. Pull on the bent elbow until you feel tension



Keeping the shoulder back, gently pull the bent arm across the chest.

in the shoulder and upper back. Hold the stretch for ten seconds. Release the stretch for a few seconds before repeating for twenty seconds more. Repeat the stretch with the other arm.



ROTATOR CUFF STRETCH

The towel (or dowel) stretch is a great exercise to help improve the flexibility and range of motion of your shoulder, and it is a must-do before every climbing and training session. Specifically, this



Hold a towel or dowel behind your head and grasp the other end behind your back. Gently pull the dowel (or towel) upward to stretch the rotator cuff muscles.

exercise involves internal rotation and shoulder extension of the arm in the bottom position (thus stretching the external rotators) while it places the other (top) arm in external rotation and shoulder flexion. To do it, roll up a towel and hold one end above and just behind your head, and then grab the other end of the towel with the opposite hand (palm facing backward) behind your back. Pull up on the towel with the top hand until you feel the light stretch; hold this position for ten to twenty seconds. Perform the stretch two or three times on each side.



ANTERIOR SHOULDER, CHEST, AND BICEPS STRETCH

This exercise provides a global stretch to improve shoulder mobility and, specifically, shoulder extension. This stretch is my personal favorite for the upper body because it addresses all my common tight spots—the biceps, pectorals, and shoulders. Sit on the floor with feet flat and knees bent about 90 degrees. Position your arms just behind your hips with the elbows straight, palms flat on the floor, and fingers pointing back. Slowly walk your hands away from your hips until you feel mild tension in your shoulders, pecs, and biceps. Be sure to keep your shoulders back (do not let them roll forward)—thinking about extending your lumbar spine and pushing your chest forward will provide



Walk your hands backward while keeping the lower back straight, chest out, and shoulders back.

a nice stretch into the pectorals. Hold this position for twenty seconds. Walk your fingers back a bit farther to enhance the stretch. Hold this position for twenty to thirty seconds before releasing the stretch.



LATISSIMUS, SHOULDER, AND TRICEPS STRETCH

This stretch will improve range of motion and shoulder mobility in the overhead, externally rotated arm position common to most climbing movements. Stand erect with arms overhead and bent at the elbows. Grab one elbow and gently pull it toward the back of your head until you feel a stretch in the back of the upper arm. Hold the stretch for ten seconds. Release the stretch for a few seconds and then perform a secondary stretch for about twenty seconds. You can extend this stretch down through the latissimus dorsi muscle by leaning slightly sideways in the direction of the pull. Repeat the stretch with the other arm.



Pull the elbow toward the back of your head until you feel tension in the shoulder and lats.

Tips for Safe and Effective Flexibility Training

1. Always engage in five to fifteen minutes of jogging, jumping jacks, easy climbing, or foam rolling before beginning flexibility training. Stretching a cold muscle can lead to injury.
2. Stretch in a slow, gradually progressive manner. Stretching should produce mild discomfort, but never sharp pain.
3. Perform a primary stretch of ten to thirty seconds. Release the stretch for a few seconds before performing a secondary stretch of twenty to thirty seconds.
4. Direct slow, deep breathing throughout the stretch. Inhale through your nose and exhale through your mouth.
5. Maintain a neutral back position—neither rounded nor hyperextended—to maximize the stretch and avoid injury.
6. Limit “gain” stretching to the legs and hips. Be conservative in stretching the shoulders.
7. Using foam rolling in conjunction with stretching will have a synergistic effect. Roll first, stretch second.
8. Refrain from excessive stretching of the forearm flexors prior to climbing, since this may affect the nervous system in a way that reduces your maximum strength and power for up to one hour. Favor light stretching, Armaid use, and sports massage prior to performance climbing.

Lower-Torso and Leg Stretches

The legs, hips, and lower back are areas where many climbers benefit from improved flexibility. Increased hip turnout, wider stemming, and better ease at high-stepping will markedly improve center-of-mass positioning and quality of movement—the upshot, of course, is a conservation of energy. While your ultimate degree of flexibility is largely a function of genetics, dedicated daily stretching and mobility work will produce some gains in functional range of motion. Consequently, engaging in a modest amount of lower-body stretching is a smart thing to do, not only as part of a pre-performance warm-up, but also as part of some before-bed mobility work and rest-day flexibility training. Here are nine stretches to put to use beginning today.



BUTTOCKS AND LOWER-BACK STRETCH

This stretch will improve hip flexion and help facilitate high-stepping. Lie flat on your back with both legs straight. Bend one leg and grasp it behind the thigh or around the upper shin and pull it toward your chest. Hold the stretch for ten seconds and



Pull the bent leg toward your chest and hold.

then release it for a few seconds. Pull the bent leg toward your chest again for a secondary stretch of about twenty seconds. Repeat with the other leg.



HAMSTRINGS STRETCH

The hamstring muscles, along the back of the thigh, are chronically tight in many climbers, thus restricting stem and high-step movements. Lie flat on your back with one leg straight and the other bent with the foot flat on the floor. Lift the straight leg upward, grab it behind the thigh or calf, and pull gently forward until you feel the stretch down the back of the leg. (Alternatively, you can keep the



Pull the straight leg forward and hold.

non-stretching leg flat on the floor as shown in the photo.) Hold this stretch for ten seconds and then release it slightly for a few seconds. Pull the leg once again for another twenty to thirty seconds. Be sure to maintain a straight leg throughout, and perform this stretch equally on both legs. You can also work this stretch by looping a fitness band or towel over your foot and regulating the stretch by pulling on the band.



ADDUCTOR STRETCH

This wall stretch is one of the very best lower-body stretches for climbers. Lying on the floor eliminates strain on the lower back and allows you to relax and



Separate legs, relax, and let gravity extend the split.

let gravity do the work. Wearing socks will reduce friction between your heels and the wall, helping maximize the range of the stretch. Lie on the floor with your buttocks about six inches from a wall and your legs extending straight up it with about a 90-degree bend at your hips. Slowly separate your legs by sliding your heels out to the sides. Concentrate on relaxing throughout your body, and allow gravity to extend the split until you feel mild tension in your legs and groin. Hold this position for twenty to thirty seconds. Try to split your legs farther apart; if needed, press on your thighs to apply some downward pressure to extend the stretch. Hold this position for a minute or two.



GROIN STRETCH

This is an excellent stretch for improving hip turnout. Flexibility gains from this stretch will allow you to move your center of mass in closer to the wall—



- 1. Gently press elbows down on thighs.***
- 2. Lie flat and allow gravity to pull knees downward.***

more over your feet—on near-vertical climbs. Sit upright with your legs flexed and knees out to the sides so that you can bring the soles of your feet together. Grasp your ankles and rest your elbows on the inside of the thighs. Press down with your elbows to apply light pressure on both thighs until you feel mild tension in your groin and inner thigh. Hold this stretch for ten seconds and then release it for a few seconds. Apply pressure for a secondary stretch of twenty to thirty seconds. Next, lie flat on your back while keeping your feet together. Relax and allow gravity to pull your knees toward the floor for another minute or two. To enhance the stretch, have a partner apply light downward pressure on your knees.



PIRIFORMIS

The piriformis muscle lies deep beneath the gluteal muscles and assists in lifting and rotating the thigh laterally. In climbing, this muscle is used to position your foot on a hold that's up and out to the side. If you have trouble reaching out your foot to engage distant edges, a tight piriformis is likely part of the problem. The piriformis stretch may also provide some relief for individuals experiencing mild sciatica. Here's how to stretch it.



Gently lower your hips and torso toward the bent leg.

Kneel on the floor with your hands positioned under your shoulders. Shift your upper-body weight onto your arms so that you can position one leg in front of you with the knee acutely flexed so that the outside of the foot contacts the floor near the opposite hip. Slide the other leg backward until it's nearly straight with the kneecap and toes touching the floor. Maintaining a straight torso, lower your hips toward the floor to increase the stretch. Hold this position for ten to twenty seconds and then raise your hips slightly to release the stretch for a few seconds. Lower your hips again for a secondary stretch of thirty to sixty seconds; repeat the stretch with the other leg. **Warning:** Perform this stretch with caution, and ease up at the first sign of any pain in the knees, hips, or lower back.



HIP FLEXORS AND QUADRICEPS

The hip flexor muscle group is commonly tight, not only because of the critical role they serve in running, hiking uphill, and lifting the legs in climbing, but also because they can readily shorten during extended periods of sitting. This makes the hip flexors, including the powerhouse psoas, a prime muscle group to target with foam rolling and stretching. Here's how to do it.

Kneel on the floor with one leg forward and bent at near 90 degrees, while the other leg is nearly straight and extended behind you. Resting your hands on the knee for balance, shift your hips forward while maintaining a flat or slightly extended spine—this should create some tension in the hip flexors atop the straight leg. Hold this position for ten seconds and then raise your hips slightly to release the stretch for a few seconds. Now, shift your hips forward again and, if you can maintain balance, extend your arms straight overhead. This will enhance the stretch and perhaps elicit a “pop” from the lumbar zone. Hold the stretch for thirty seconds. Release the stretch and then flex your rear leg so that you can grab it with the opposite hand.



- 1. Shift hips forward until you feel tension in the front of your hip.**
- 2. Pull foot upward with opposite hand to stretch the quadriceps.**

Now do a final stretch in which you both shift forward and pull gently on the rear foot. This will provide an excellent stretch of the quadriceps.



CALF STRETCH

There are many ways to stretch the calf muscles of the lower leg, so there's no reason not to give these hard-working muscles a few stretches throughout the day. By elevating the front part of the foot, on a small step or other object, and keeping your heel on the floor, you simply need to lower your hip level and flex your knee forward to elicit a good stretch. Hold the stretch for twenty seconds, release it for a few seconds, and then resume the stretch for another thirty seconds. Repeat the stretch with the other leg.



Elevate your forefoot and shift your knee forward while maintaining heel contact with the ground.



ABDOMINAL STRETCH

While strong abdominals that can stiffen on demand are highly coveted, it's equally important to be able to quickly relax the abs and extend the torso. Most people find it relaxing and pleasurable to stretch the abdominals and extend the spine before and following a workout. Begin in a prone position with your arms bent and palms flat on the floor next to your shoulders. Slowly press your shoulders away from the floor until you feel mild tension in your abdominals. Keep your legs and pelvis in contact with the floor for the duration of this



Keeping your hips and legs grounded, push your shoulders away from the floor until you feel a stretch in the rectus abdominus.

stretch—it helps to contract your buttocks in order to maintain the position and reduce stress on your lower back. Hold the stretch for ten seconds and then return to the starting position for a few seconds. Press up for a secondary stretch of twenty to thirty seconds. Relax and allow the curve of your spine to extend up through your upper back and neck. Look forward, but not up toward the ceiling. Do not be overly aggressive with this stretch—proceed with caution, especially if you have a history of back problems. Stop immediately if you experience any pain in the lower back.



OBLIQUES, HIPS, AND BACK STRETCH

This final global stretch—of the hips, torso, and back—will help prepare your body for the torso-twisting, hip-turning, drop-kneeing movements that are so common on steep sport climbs and boulder problems. Sit erect on the floor with one leg straight and the other bent and crossing over the opposite knee. Slowly turn your body toward the side of the bent leg until you feel mild tension in your lower back, hips, and the side of your torso. Maintain a level head position and fix your eyes on the wall to the side of the bent leg. Hold the stretch for ten to twenty seconds and then return



Gently rotate your torso toward the bent-leg side.

to the starting position for a few seconds. Perform a secondary stretch for twenty to thirty seconds. If needed, you can increase the stretch by leveraging your elbow against the thigh of the bent leg. Repeat the stretch in the other direction. Be sure to switch leg positions.

Stabilizer and Antagonist Muscle Training

This section on training the stabilizer and antagonist muscles may be the single most important in the entire book. While few climbers need additional motivation to train their finger or core strength, it's my experience that the majority of climbers spend too little time training the crucial muscles that support the wrist and shoulder during strenuous climbing moves. Left untrained, these “supporting cast” muscles, which stabilize the wrist and scapula, will eventually let you down in critical moments of fatigue—the result could be insidious slow-onset elbow tendinitis or shoulder impingement, or a more acute rotator cuff tear.

As common as injuries are in this sport, it surprises me that so few climbers are dedicated to regular training of the antagonist muscles. No doubt, training these smaller muscle groups lacks

the excitement of bouldering or campus training, and low-information climbers might view it as unnecessary, boring, or a waste of time. It's my opinion, however, that every climber should train their stabilizers and antagonists twice per week. Furthermore, the harder a person trains and climbs, the more indispensable these exercises become for preventing injury and ensuring longevity in the sport. So shelve the excuses, and make the commitment today. Coach Hörst insists!

The three subsections that follow will provide rich details on training the wrist stabilizers, the scapular stabilizers and rotator cuff, and the larger push muscles that oppose and provide balance to the many climbing-specific push muscles. (Consult the anatomy photos in appendix A if you are unfamiliar with any of the muscle groups mentioned.)

Training the Wrist Stabilizers

The musculature of the forearms is some of the most complex in the body. For example, at first blush it seems that the forearm flexor muscles are almost single-handedly (pun intended) responsible for securing a tight grip to the rock. Consequently, exercises that target the forearm flexor muscles—such as the fingerboard, HIT System, campus board, or others—are the focus of most climbers' training programs. The problem is that developing strong fingers and forearms that are also resistant to injury demands that you also regularly train the muscles that extend the fingers and wrist. Here's why.

The finger flexors suffer from a sort of functional insufficiency in that they are much weaker when the wrist is flexed and the muscles are short. As an experiment, flex your wrist as much as possible (wrist angle around 90 degrees) and pinch your middle finger as hard as possible against the thumb. It's hard to create much force, right? Now, while still pressing your middle finger against the thumb, slowly extend your wrist to a straight position and then an over-extended position. What did you notice? If you tried to squeeze hard throughout this

range of motion, you likely discovered that your finger strength increases as the wrist extends.

Biomechanically, there are a couple of reasons that finger strength is greater with an extended wrist position. First, muscles can contract more forcefully when they are elongated, as the finger flexors are with an extended wrist position. More interesting, however, is how the fingers naturally assume a hook-like position when the wrist is over-extended. Try it right now—over-extend your wrist and you'll see the meat hooks revealed!

Now you know why fatiguing climbers lift their elbows into the chicken-wing position—it puts their wrist into a position that enables their pumped forearms to grip a little longer. Done chronically, however, such chicken winging stresses the extensor muscles and puts added stress on the tendon pulleys in the fingers. The result can be gradual onset of lateral epicondylitis and/or finger tendon problems.

The training “fix,” then, is to dedicate a modest amount of time to training the finger/wrist extensor muscles. No, you can't just extend your fingers against a rubber band and call it good. You need to do some serious training in multiple ways, just as you train the finger flexors in a comprehensive manner! It's paramount to train the wrist in both a neutral and extended position, as well as in a wide pinch grip position—this is the single most overlooked and important position to train it. While you don't need to do all the exercises every workout, at minimum I suggest doing reverse wrist curls, wide pinches, and pronator isolation. Do one set of these three exercises (with a light resistance) as part of your warm-up for climbing (or training), and then do two more sets with higher resistance toward the end of your workout.



FINGER EXTENSION AGAINST RUBBER BAND

Okay, let me get this one out of the way first. Use of a rubber band (or similar) to extend your fingers

Powerfingers are an excellent tool for warming up and rehabbing the extensors.



against is sufficient only for the purposes of warm-up and rehabilitation from injury. I recommend use of Powerfingers to warm up before advancing to the exercises below.



REVERSE WRIST CURLS

Just as the name implies, this exercise involves curling a dumbbell upward with your palm facing down. Fix your forearm in a stable, horizontal position using a bench, table, or on your thigh (if sitting) and grasp the dumbbell with your hand overhanging the supporting surface. Begin with the dumbbell in the top (over-extended) position and lower it to the neutral (straight wrist) position, then curl the dumbbell back to the top position. Continue with slow, controlled reverse curls at a rate of about one rep per second. I recommend doing two sets with progressive resistance. For the first set select a weight that allows twenty to twenty-five repetitions, but stop the exercise before complete failure. Rest three minutes before doing a second set with a heavier dumbbell that allows only ten to fifteen reps. Initially you may need to use only a five-pound dumbbell; advanced climbers may need as much as a twenty-five- to thirty-pound dumbbell for the high-resistance set. Do some



- 1. Start in the top position.***
- 2. Stop at neutral position and return to top position.***

mild stretching of the extensors during each rest period.

Note: Anyone with elbow pain or recent history of lateral tendinosis should do only the negative phase of this exercise—that is, use your free hand to grab the dumbbell-holding hand and aid its return to the top position after each negative (lowering phase).



WRIST EXTENSION ISOMETRIC

I love this exercise! You'll need a clock (with a second hand) in view or a timing app to train against. The setup is the same as in the reverse wrist curl, but you'll be using a lighter-weight dumbbell. The difference here is that there's no lowering phase—you'll simply hold the dumbbell in the up (over-extended position) for a long isometric contraction. Select a dumbbell light enough to allow an isometric hold in the over-extended position for forty-five to sixty seconds. Do two sets with each arm with a three-minute rest in between. Long term, strive to hold the over-extended position for a full two minutes. It'll feel good to do some stretching of the forearm extensors during the rest period.



WIDE PINCH WITH WRIST EXTENSION

This novel exercise is absolutely essential, as it strengthens the extensor muscles with the fingers fully extended to mimic grabbing open-hand and wide pinch holds in climbing. It may seem like a small distinction, but the wrist extensors function a bit differently when the fingers are straight (extended) compared with when the fingers are flexed, as in crimping or holding a dumbbell. So it's important to do this exercise in addition to one of the previous.

Until there's a commercially available device for training the wide pinch grip, you'll need to kludge something that will work. Screwing together two or three pieces of 2x4 wood blocks is a cheap,



Wide pinch with wrist extension.

1. Using a bumper plate.

2. Using wood blocks.

effective solution—consider making two sizes, wide and extra-wide. Another option is pinching a thick bumper weight plate. Anyway, the exercise is straightforward: Standing upright with good posture, pinch the wood block or bumper plate with a straight arm and extended wrist; hold this position for ten to thirty seconds.

Initially, I suggest training for endurance, which will require a light weight that allows a full thirty-second hold. Do three with each hand with at least a minute rest in between. Longer term, consider using a heavier weight that allows only a ten-second hold. In this case, do three consecutive ten-second pinches with the same hand, resting only thirty seconds between each one. Do a total of three sets of three reps with each hand, resting for about three minutes between sets.



PRONATOR ISOLATION

Arm-pulling movements naturally result in supination of the hand. If you perform a pull-up on a free-hanging set of Pump Rocks, you'll discover that your hands naturally turn inward, or supinate, as your biceps contract. Consequently, training forearm pronation is an important antagonist exercise for climbers to maintain muscle balance across the forearms. While a well-equipped gym may have a machine for training forearm pronation, most folks will need to find a creative training solution. My preference is using an ordinary three-pound sledgehammer!

Sit on a chair or bench with your forearm resting on your thigh, hand in the palm-up position. Firmly grip the sledgehammer with the heavy end extending to the outside and the handle parallel to the floor. Turn your hand inward (pronation) to lift the hammer to the vertical position. Stop here. Now slowly lower the hammer back to the starting position. Stop at the horizontal position for one second before beginning the next repetition. Continue lifting the hammer in this way for twenty to twenty-five repetitions. Choke up on the hammer if this feels overly difficult. Perform two sets with each hand.

Alternatively, you can cut a sixteen-inch length of one-inch-diameter dowel and mount a two-and-one-half pound weight on one end.



1. Palm up. 2. Lift hammer to vertical.



PRONATOR ISOLATION WITH AN EXERCISE BAND

As an alternative to the sledgehammer exercise above, you can train the pronator with an exercise band. In a sitting position, anchor one end of the band under your foot and then wrap the other end of the band around your hand. Remove all the slack between your hand and foot so that there's tension when holding the band palm up with the back of your forearm resting on your thigh. Keeping your forearm in a fixed position, turn your hand inward (pronation) to reach the palms-down position. Return to the starting position and continue for twenty-five repetitions. Make sure there's enough



1. Anchor one end of the band under your foot and grasp the band palm up. 2. Rotate your forearm inward until palm is facing down.

tension in the band to provide moderate resistance throughout the full 180-degree range of motion. Do two sets with each hand.



REVERSE ARM CURLS

The reverse arm curl is one of my favorite warm-up exercises. While it obviously activates the biceps, more importantly, it demands an isometric contraction of the wrist extensors and calls into play the often overlooked brachioradialis to stabilize and flex the elbow. In elbow flexion the brachioradialis is most active when the hand is pronated, as in the majority of climbing moves—therefore



With a palms-down grip (pronated forearms), curl the barbell upward while keeping your elbows fixed.

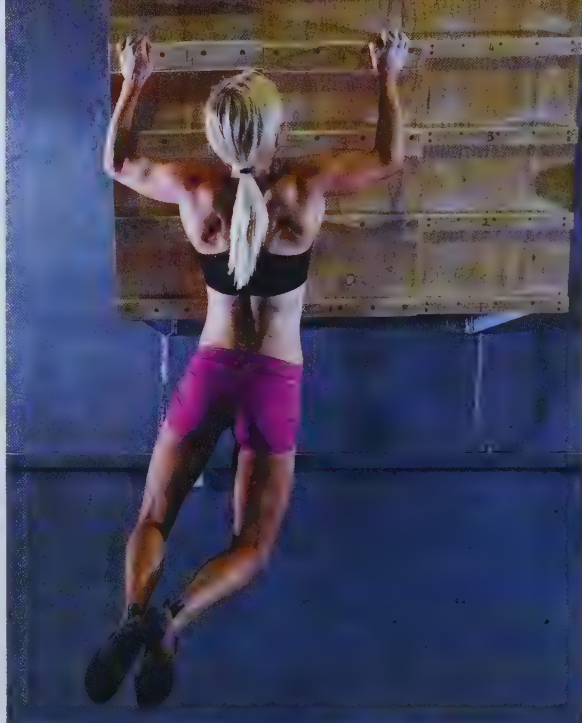
strengthening the brachioradialis means a stronger wrist and a more stable elbow and forearm while gripping and pulling down on the rock.

Do this exercise with a barbell or bent E-Z curl bar (see photo), not with dumbbells. Take an overhand grip and, with elbows fixed at your side, curl the bar upward to reach full elbow flexion. Continue for fifteen to twenty repetitions, being sure to contract your core, glutes, and spinal erectors throughout the duration of the exercise. I suggest doing one set with light weight as part of your warm-up, and another moderately heavy set or two with your antagonist training.

Training Rotator Cuff and Scapular Stability

The rotator cuff is arguably the most stressed group of muscles in the body of a climber. While the forearms get all the attention—and pump—the rotator cuff and scapular stabilizers are the real heroes as a climber repeatedly pulls with his arms from a wide range of angles, both statically and dynamically applying force of varying intensity. Climbing on overhanging walls, at times with jumping, lunging, and campusing moves, challenges the small rotator cuff muscles to keep the humeral head in place; similarly the scapular stabilizers must work hard to move the scapula into the proper position for each given arm position in climbing. Even in vertical climbing the crimp grip and the chicken-wing arm position stress the rotator cuff, and, not surprisingly, shoulder pathology will weaken your grip. Hence, dysfunction anywhere along the chain of force application—fingers, wrist, elbow, or shoulder—will negatively affect performance of the entire system.

If you still aren't motivated to train the cuff and scap, roll this over in your mind: Weak rotator cuff muscles are a common, yet unrecognized, limiting constraint in your maximum pulling strength, lunging power, grip strength, and contact strength. So if you're a hard-training climber frustrated by a lack of strength/power gains on the rock, it may be that your central governor is limiting power output



Well-developed middle back muscles, as Anna Davey displays here, are essential to stabilize the scapula during strenuous arm movements common to hard climbing and campus training.

THERON KIRKMAN

due to afferent feedback from your weak, unstable shoulder joints. This is why it's essential to develop stability before strength, and strength before power (per figure 6.1).

Anyway, I trust you'll agree that the shoulder is a truly wondrous and remarkably dynamic joint, and therefore it's essential that you are proactive in training (and addressing weaknesses in) the many muscles involved. By using both the aforementioned mobility exercises and the strength exercises that follow, you will improve movement patterns and function, even when climbing in a state of high fatigue (when injury risk is highest). First, I'll detail two exercises that isolate the four muscles that comprise the rotator cuff: the supraspinatus, infraspinatus, teres minor, and the subscapularis. Next, I'll reveal four exercises that work the supporting cast of muscles, including the upper, middle, and lower trapezius, rhomboids, and serratus anterior, that move the scapula into

proper position for overhead arm movements. Then, in the next section (antagonist training) I'll present three pushing exercises that work the entire system, including the larger role players of the pectorals, deltoids, and latissimus.



DUMBBELL INTERNAL ROTATION

Of the four rotator cuff muscles, only the subscapularis contributes to internal rotation. It does get substantial help, however, from the deltoid, teres major, latissimus, and pectorals—this helps explain why many climbers find they are stronger at internal rotation than external rotation. Still, you want to do this exercise twice per week since it (partially) isolates and strengthens the subscapularis.

Lie on your side with your bottom arm in front of your waist; place a rolled-up towel under your head to support your neck. Rest your other arm along your hip and upper thigh. Hold a five- to fifteen-pound dumbbell in the hand of your bottom arm, positioning this forearm perpendicular to your body. Lift the weight up to your body and hold for a moment before lowering it back to the floor. The upper portion of your arm should remain fixed throughout the range of motion—think of the upper arm and shoulder as a door hinge that allows your forearm to swing “up and down.”



1. Starting position. 2. Ending position.

Continue in a slow but steady motion for a total of twenty to twenty-five repetitions. Do two sets on each side, with a three-minute rest between sets. Select a weight heavy enough to make you work, but not so great that you have to strain hard or compromise technique. Increase weight in two- to five-pound increments, as needed.

An alternative method of training the internal rotators is to anchor an exercise band to a waist-high object and, while standing, grip the band and pull it across your body. The key is to keep your elbow and upper arm in a fixed position and only move your hand and forearm across your body. Repeat this twenty-five times, adjusting resistance as needed by changing your distance from the anchor point or by using a different exercise band. In my opinion, such exercise band training, while good for rehab, doesn't cut it for serious training of the rotator cuff. Therefore I recommend using the dumbbell exercises detailed here.



DUMBBELL EXTERNAL ROTATION

The infraspinatus and teres minor are the primary external rotators, although they get some help from the deltoid. Many climbers are surprised to discover that they have weak external rotators and thus may be able to use only about one-half as much weight as when doing the internal rotation exercise above. Strengthening the external rotators should be a high priority, as this will provide additional protection of the cuff when grabbing the rock with the stressful above-the-head Gaston positions common to hard climbing.

Lie on your side with your bottom arm in front of your waist and a rolled-up towel under your head to support your neck. Alternatively, you can bend your bottom arm and use it as a headrest. Hold a five- to ten-pound dumbbell in the hand of your top arm. Rest the upper arm and elbow on the top side of your body, and then bend at the elbow so that the forearm hangs down over your belly and the weight rests on the floor. Keeping



1. Starting position. 2. Ending position.

your elbow and upper arm fixed, lift the weight upward toward the ceiling and stop when you feel the resistance ease as you move past the horizontal position. Return the weight to the starting position and continue for twenty to twenty-five repetitions. Do two sets on each side with a three-minute rest between each set. Increase weight in two- to five-pound increments. Initially you may need to use as little as five pounds. With long-term training, however, you should be able to improve to using a ten-pound dumbbell; especially fit climbers may be able to use up to fifteen pounds. This is a difficult exercise, and using proper technique is essential for isolating the external rotators. Err on the side of using too little weight, rather than too much.

If you don't have access to dumbbells, you can use an exercise band as an alternative, though perhaps less effective, training method. With an exercise band anchored to a waist-high object, stand with your side (opposite the arm to be trained) facing the band anchor spot. Grab the free end of the band with your forearm across your belly and upper arm at your side. Keeping your elbow and upper arm in a fixed position, pull outward on the exercise band with your forearm opening up away from the body like a swinging door. Repeat this twenty-five times, adjusting resistance as needed by changing your distance from the anchor point or using a different exercise band.



"T" EXERCISE

This exercise targets the middle trapezius and rhomboids, the muscles that produce scapular retraction. You can do the Ts in a standing position



Ts with TRX Sling trainer.

1. Starting position. 2. Finish position.

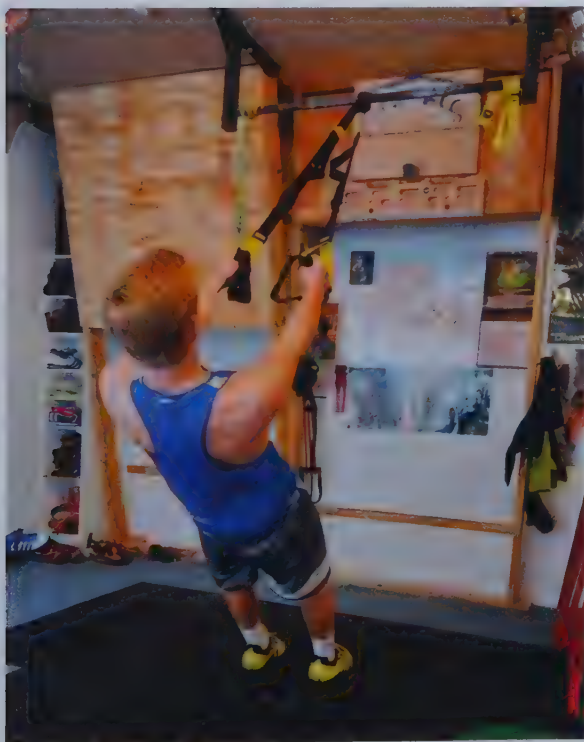
with a TRX trainer or prone on a bench. Using the TRX, grasp the handles with straight arms extended forward and palms facing each other. Now contract your core, glutes, and leg muscles and lean backward to weight the TRX; you can also walk your feet slightly forward (farther forward is harder). Maintaining a tight torso and straight arms, pull your hands apart until your arms are straight out to your side in a T position—think about squeezing your shoulder blades as close together as possible. Slowly lower back to the starting position and continue for ten to twenty repetitions. Do two sets. If you don't own a TRX trainer, you can still train Ts in an effective way by assuming a prone position on a bench with your arms hanging straight down. Contract your core and glutes, and lift straight arms out to the side until raised just above horizontal. Feel your shoulder blades come together, and

actively squeeze them at the top position. Continue for twenty repetitions. You can progress to using very light dumbbells (a few pounds max).



“Y” EXERCISE

This exercise is only slightly different from the Ts described above, but the Y motion specifically targets the lower trapezius. The lower traps produce scapular depression and upward rotation, which is essential to avoid shoulder impingement when doing overhead movements. Among climbers, the lower traps are commonly a weak link and thus a potential contributing factor to shoulder injury. If you frequently engage in campus training and lunge moves, then a strong mid and lower trapezius are vital to facilitate proper scapular positioning (to avoid trashing your shoulders).



Ys with TRX trainer.

1. Starting position. 2. Ending position.

Set up in the same way as when doing Ts, but the motion is up and outward to form a “Y.” Stop pulling when your arms come in line with your body—at this top position your arms should be just beyond shoulder width apart. Be sure to contract your core, glutes, and legs to maintain rigidity throughout the range of motion. Think about drawing your scapula downward as you reach the top of the Y position. You can also do the Y exercise lying prone on a bench with just arm weight or a one- or two-pound dumbbell in each hand. Begin with your arms straight and hanging down under the bench below your face. With a stiff core and tight glutes, lift straight arms up into the Y position with your hands just beyond shoulder width at the top position. No matter which method of Y training you select, do two sets of ten to twenty repetitions with a three-minute rest in between.



SCAPULAR PUSH-UP

The scapular push-up isolates the serratus anterior muscle, which is vital for scapular stability and protraction. You can do this simple exercise from a straight-arm push-up position or from a plank position with your elbows and forearms flat on the floor. I suggest using the plank position for learning the feel of this exercise. Assume the classic plank position with elbows under your shoulders, a neutral spine, and tight abs and spinal erectors. Keeping your torso

stiff and straight, allow your chest to sag downward between your arms. Now drive your elbows and forearms into the floor to return to the top position—feel your scapula slide around your rib cage as the serratus anterior contracts. The range of motion is only a few inches, but it’s a great isolation exercise if you do twenty or so repetitions. This movement is not meant to be heavily loaded, so there’s no need to use additional weight. Do one or two sets.



SCAPULAR PULL-UP

This last exercise is anything but the least—the scapular pull-up is perhaps the best climbing exercise that almost nobody does! Keeping your shoulders healthy and developing proper movement patterns in pulling motions demands the ability to depress, rotate, and retract the scapula. Regular use of this isolation exercise will develop better kinesthetic awareness of your scapula position and enable you to climb harder and longer with good form, despite growing fatigue. Furthermore, being able to quickly and forcibly engage the lower trapezius and lats will empower you to keep your scapula in proper position when campus training and lunging.

Strong climbers can do this exercise with full body weight, but I suggest learning with less resistance by keeping your feet on the floor and flexing your knees enough to hang with straight arms from a pull-up bar. Either way, the range of motion is



1. Begin in a fully extended push-up position with scapular protraction. 2. Maintaining straight arms, allow your chest to sag downward (scapula retraction) and immediately press back to the starting position.

only a few inches—from a full hang with somewhat shrugged shoulders, the desired action is to draw the scapula down and together. The best learning cues are to try to “bend the bar,” and think about doing a reverse shrug (i.e., shoulders drawn downward). Do this and you’ll feel your head shift backward and your chest raise upward, as your scapulas pinch together. Hold the top position for one second, then return to the starting position. Do eight to twelve reps, keeping straight arms and tight spinal erectors and glutes throughout. At first you may find this to be a difficult exercise (a sign that you’ve found a critical weakness to correct!), but resist the urge to overdo it. Add a second set to your workout only after you’ve mastered the exercise.



1. Begin from an ordinary pull-up starting position. 2. Maintaining nearly straight arms, depress and downwardly rotator your scapula by pressing down on the bar—think about pushing your chest out and trying to bend the bar.

Antagonist Training

The previous sections presented the must-do isolation exercises for strengthening your rotator cuff and scapular stabilizers. This chapter concludes with three vital multi-joint exercises that target the large climbing antagonist muscles, including the three heads of the deltoids, the pectoralis, and the triceps. The importance of doing these exercises cannot be overstated, considering the need to balance out all the pulling you do in climbing with some quality push movements. As in training the stabilizer muscles, you want to train the antagonists with a moderate resistance and relatively high number of repetitions. Keep in mind that performing a high-weight hypertrophy or maximum-strength program is unnecessary and, in fact, could be detrimental if it causes unwanted gain in muscle mass.

While the three following exercises can all be performed on standard health equipment (Cybex, Olympic weights, and so forth), I would advise against purchasing a health club membership just to gain access to the necessary machinery. Instead, I recommend a one-time investment in a few pairs of dumbbells (only \$100 to \$200)—they’ll last you a lifetime. Alternatively, you might ask the climbing gym you patronize to purchase a few dumbbells or a set of free weights for the purpose of training the antagonist, rotator cuff, and stabilizer muscles. Of course, it’s also possible to do some of these pushing exercises with a sling trainer such as the TRX.



SHOULDER PRESS

The shoulder-press motion is almost exactly opposite that of pulling up while climbing—therefore, no exercise is more central to antagonist-muscle training. Although you can execute this exercise with a common health club overhead-press machine, performing dumbbell shoulder presses provides a more complete workout, including some extra work for the cuff and scapular stabilizers.

Sit on a bench with good upright posture and feet flat on the floor. Begin with bent arms, palms



1. Beginning position. **2. End with dumbbells touching end-to-end.**

facing forward, and the dumbbells positioned just outside your shoulders. Press straight upward with your palms maintaining a forward-facing position. As your arms become straight, squeeze your hands slightly inward until the dumbbells touch end-to-end. Lower the dumbbells to the starting position. The complete repetition should take about two seconds. Continue this motion for fifteen to twenty repetitions. Strive for smooth, consistent motion throughout the entire set. Rest for three minutes and perform a second set.

Women should start with five-pound dumbbells and advance to ten- or fifteen-pounders when they can do twenty reps. Most men can begin training with fifteen- or twenty-pound dumbbells and then progress to twenty-five and thirty pounds as they are able to achieve twenty-five repetitions. Over the long term it's best not to progress beyond about 40

percent of your body weight (total weight lifted), since frequent use of heavier weights may build undesirable muscle bulk.



BENCH PRESS (OR PUSH-UPS)

The bench press is a staple exercise of power lifters and bodybuilders, but it's also useful to climbers striving to maintain stable, healthy shoulders. The key is to use only moderate resistance—begin with a total weight equal to about 30 percent of your body weight and progress up to about 75 percent (no need to go higher). For example, a 160-pound climber would begin training with two 25-pound dumbbells (50 pounds total) and progress up to



1. Beginning position. **2. End position.**

training with, at most, 50-pound dumbbells or 120 pounds with an Olympic bar.

Lie flat on a bench with bent legs and your feet flat on the floor. Using an Olympic bar or two dumbbells, begin the exercise with your hands just above chest level and palms facing your feet. If you're using a bar, your hands should be a few inches wider than your shoulders. Press straight up with a slow, steady motion. If using dumbbells, squeeze your hands together to touch the ends of the dumbbells together upon reaching the top position. Return to the starting position, pause for a moment, and then begin the next repetition. With a bar, be careful not to bounce the bar off your chest. The goal is slow, controlled movement that takes about two seconds per repetition. Continue for fifteen to twenty repetitions. Rest for three minutes before performing a second set.

As an alternative, push-ups provide a workout similar to the bench press. Begin with your hands shoulder width apart and build up to doing two sets of twenty-five repetitions. If necessary, move your hands closer together to increase training resistance. Conversely, beginners should do the push-ups with their knees on the floor until they are able to progress to the normal feet-on-floor position.



DIPS

Dips are an excellent exercise for strengthening the many muscles of the upper arms, shoulders, chest, and back. What's more, the dip motion is quite similar to the mantle move in climbing and thus provides a very sport-specific benefit! Some health clubs and gyms possess a parallel-bar setup ideal for performing dips. Alternatively, you can use the incut 90-degree corner of a kitchen counter, or set two heavy chairs in a parallel position. A set of free-floating Pump Rocks or gymnastics rings are my personal favorite, as they provide a more dynamic (and difficult) workout.

Position yourself between the parallel bars, rings, or similar apparatus. Jump up into the straight-arm

starting position with your hands drawn in near your hips. Slowly lower until your arm is bent 90 degrees—do not lower beyond this point! Immediately press back up to the starting position. Continue this up-and-down motion, with each repetition taking about two seconds. Strive to complete eight to twenty (hard) repetitions. Perform two or three sets with a three-minute rest between each set. Don't rush or bounce through this exercise, and never lower beyond a 90-degree arm bend.

If you are unable to do at least eight dips, enlist a spotter to reduce the resistance as needed so that you can reach this goal. The spotter should stand behind you and lift around your waist or, more easily, pull up on your ankles (bend your legs and cross them at the ankles to facilitate this).



- 1. Begin with a straight-arm start.**
- 2. Lower to a ninety-degree elbow angle, then immediately press back up to starting position.**



*Alex Puccio sends
Unshackled Low (V11),
Lincoln Lake, Mount
Evans, Colorado.*

CHRIS NOBLE